

Modeling of a contagious disease

Computational Astrophysics

Master in Theoretical Physics

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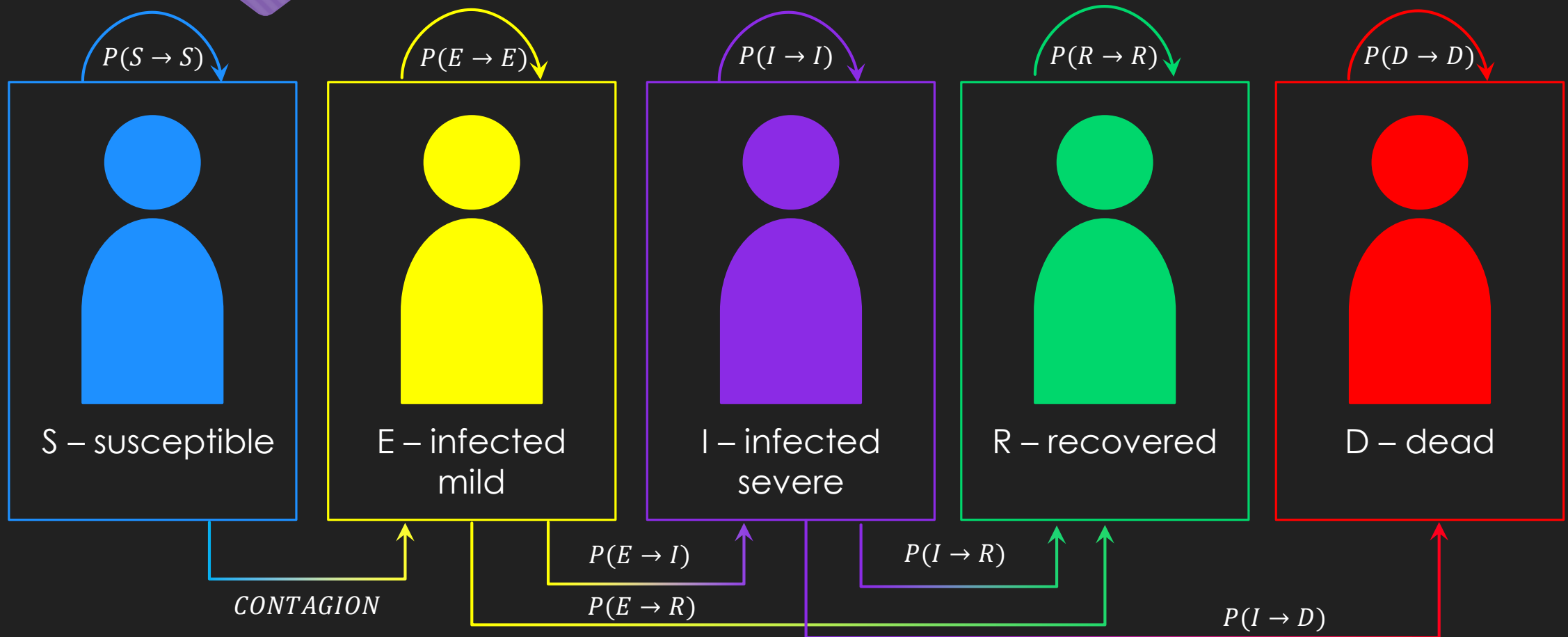
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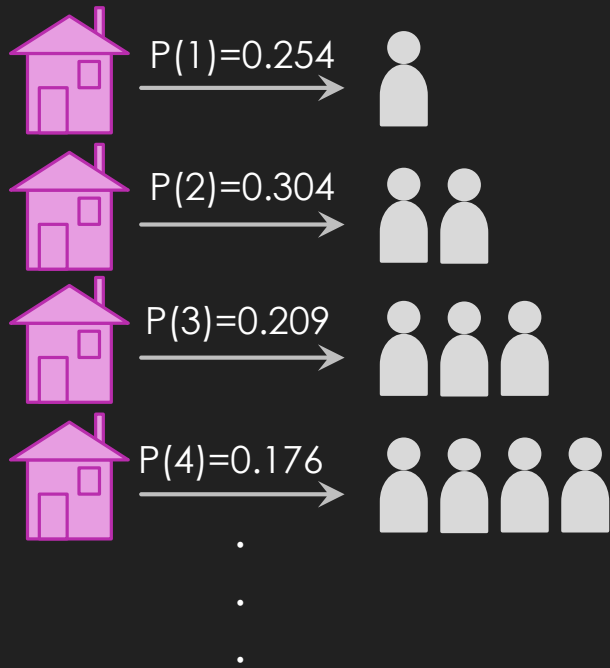
1. Introduction

- Conceptually simple problem where many C tools during this course can be applied.
- C programming and **parallelization** → larger samples of population and multiple realizations for error estimation.
- Practice creating a **modular** program with shared **structures** and **I/O**.
- Topic whose modelization can be very useful to determine and characterize the parameters to which the social evolution of the disease is more responsive.

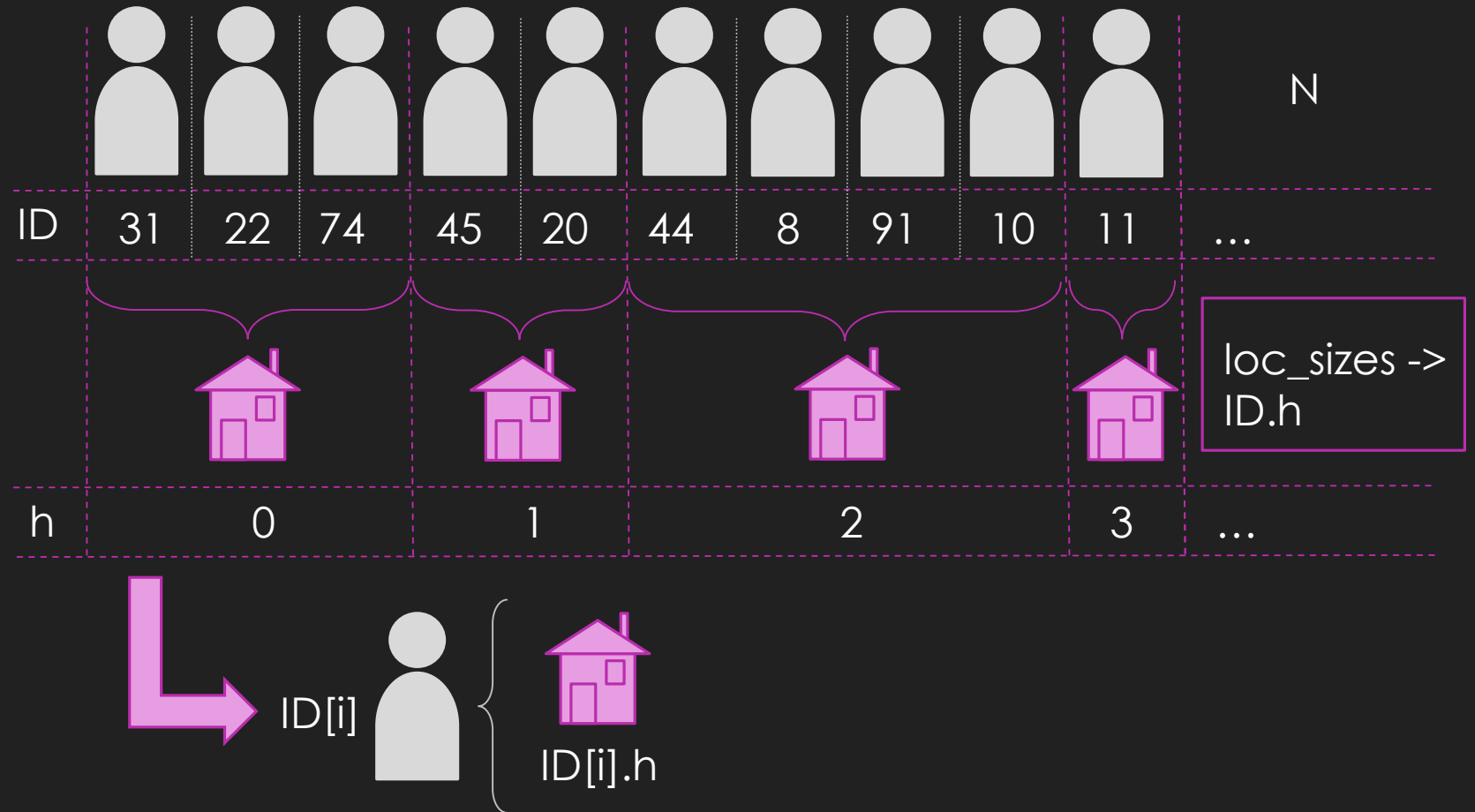
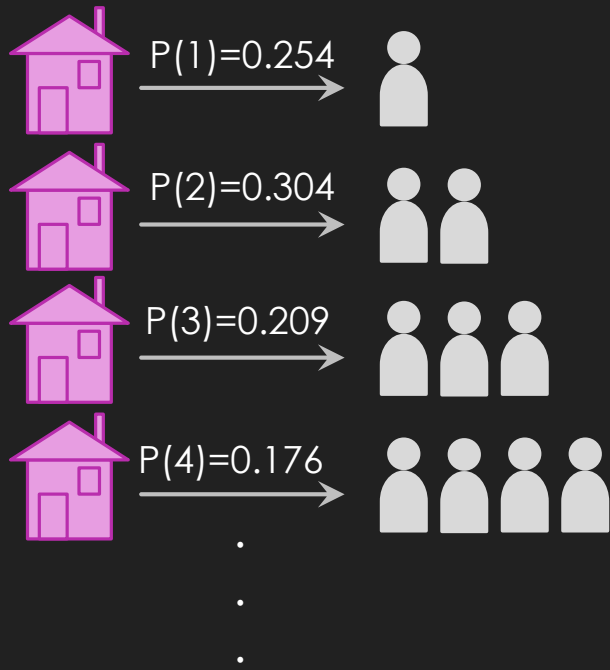
2. Implementation of a SEIRD model



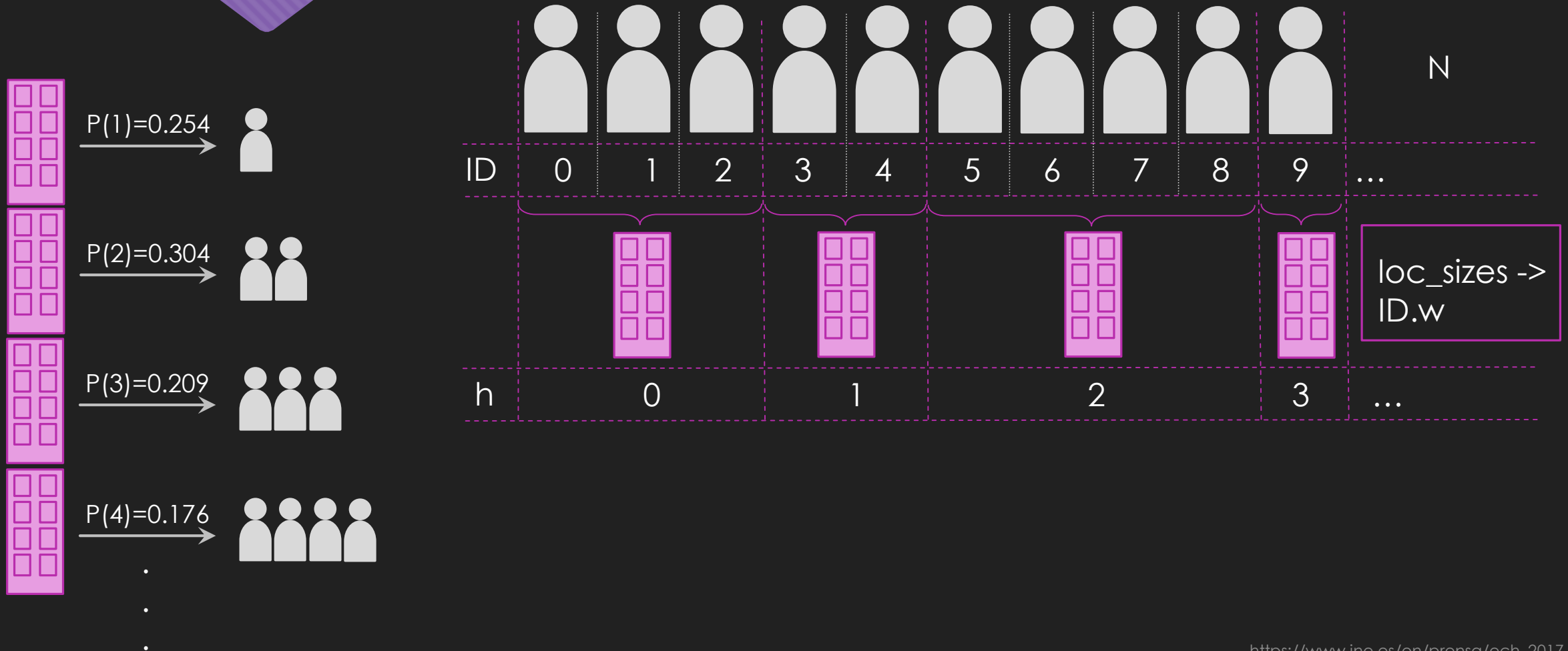
2.1 Creating social interactions



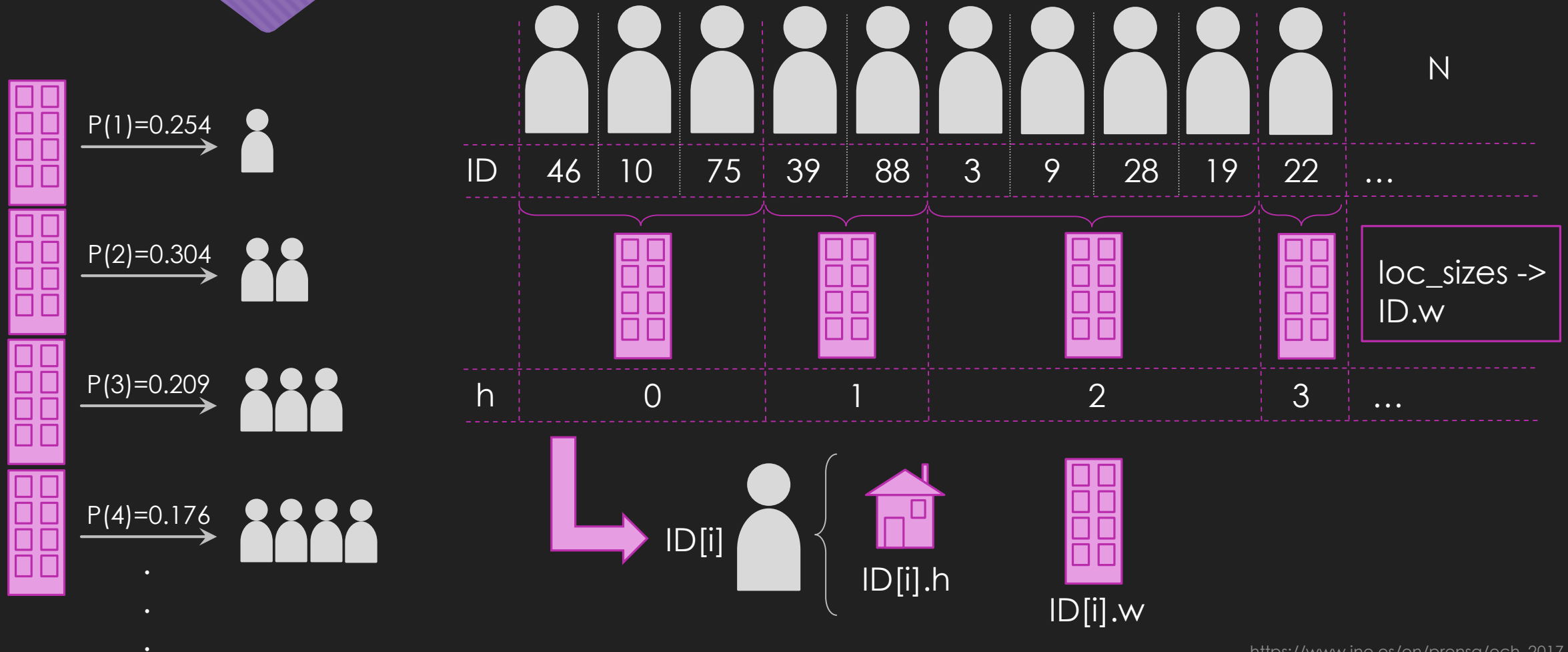
2.1 Creating social interactions



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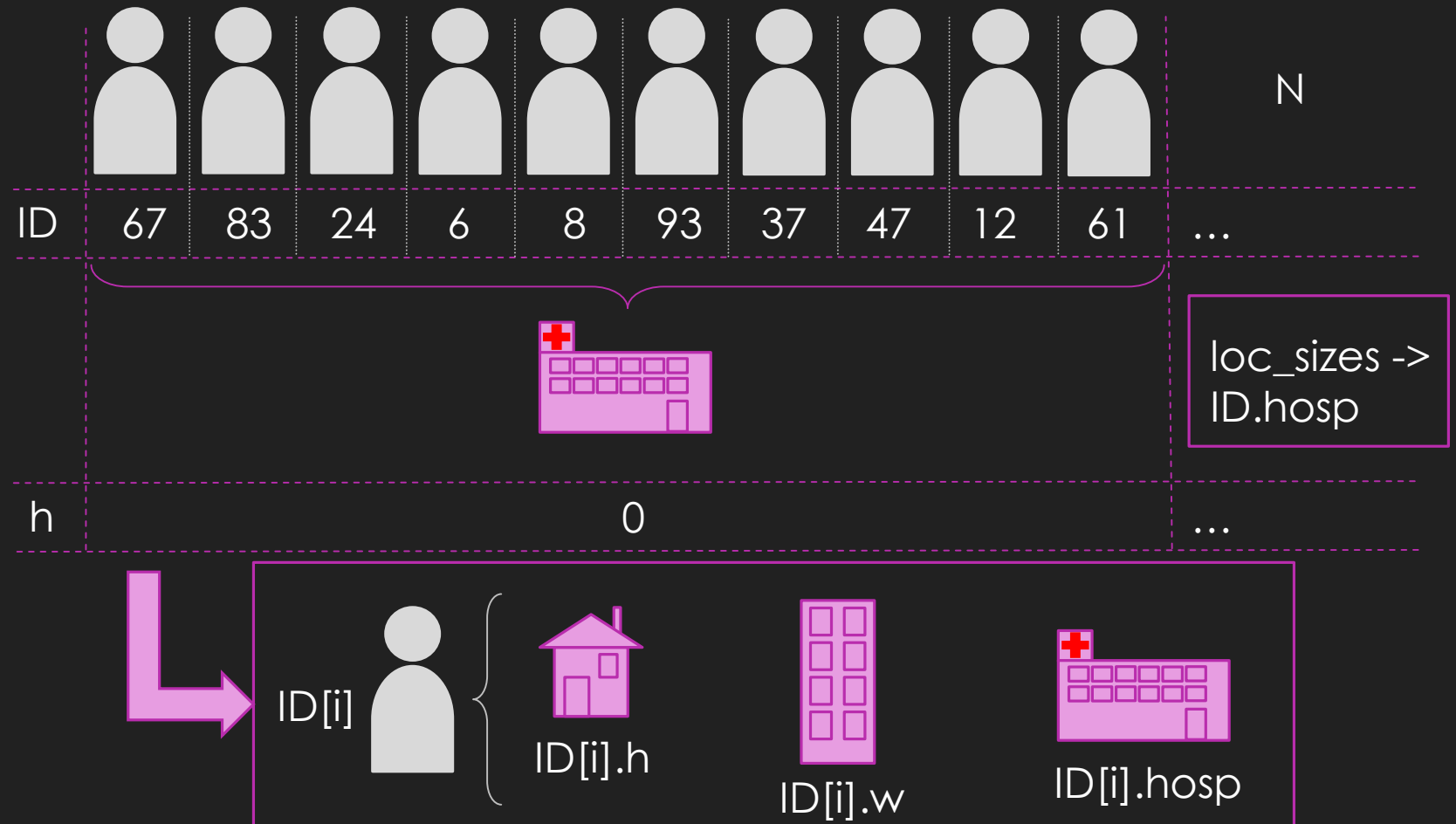
2.1 Creating social interactions



↓
× 32500

People assigned to a hospital

BUT we will assume
that when a hospital
has a certain
percentage of
occupation → ...
hospital collapse



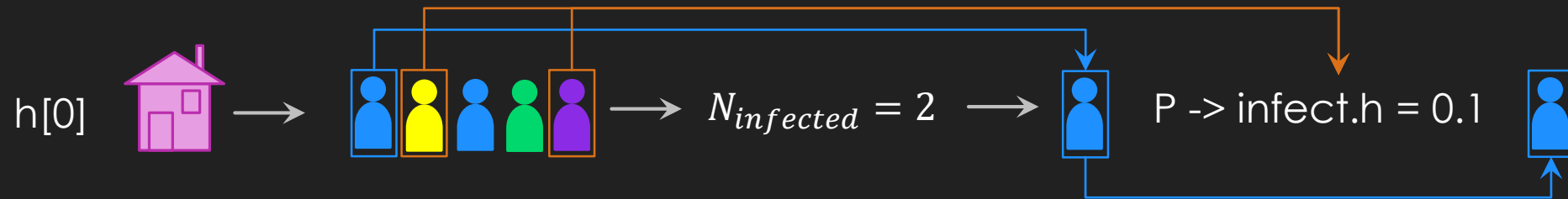
2.1 Creating social interactions

P → go[alert][state].place						
NO ALERT		1.0	1.0	0.05	1.0	0.0
		1.0	0.5	0.0	1.0	0.0
		0.005	0.05	0.95	0.005	0.0
ALERT		1.0	1.0	0.05	1.0	0.0
		0.05	0.025	0.0	0.05	0.0
		0.005	0.05	0.95	0.005	0.0

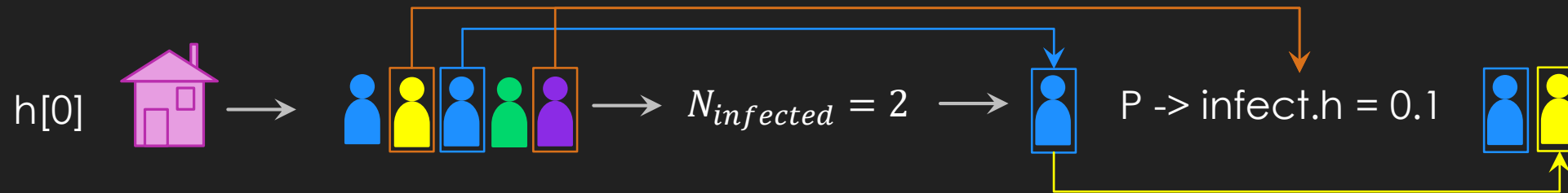
2.2 Contagion



2.2 Contagion

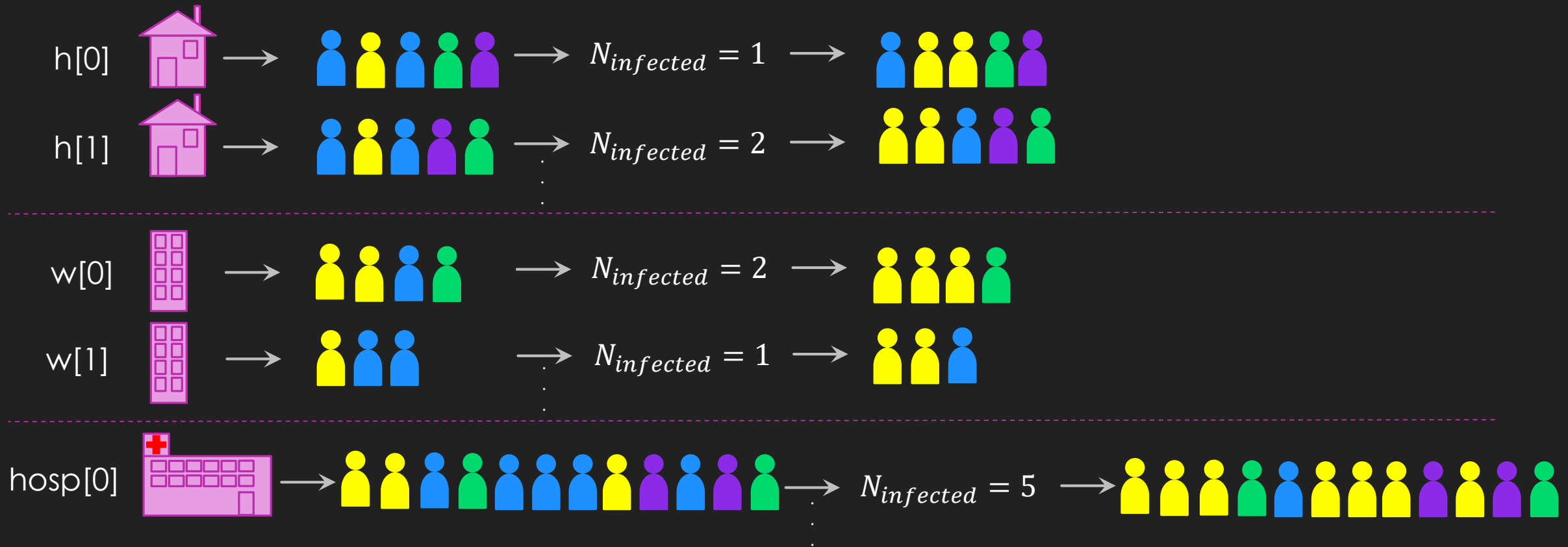


2.2 Contagion

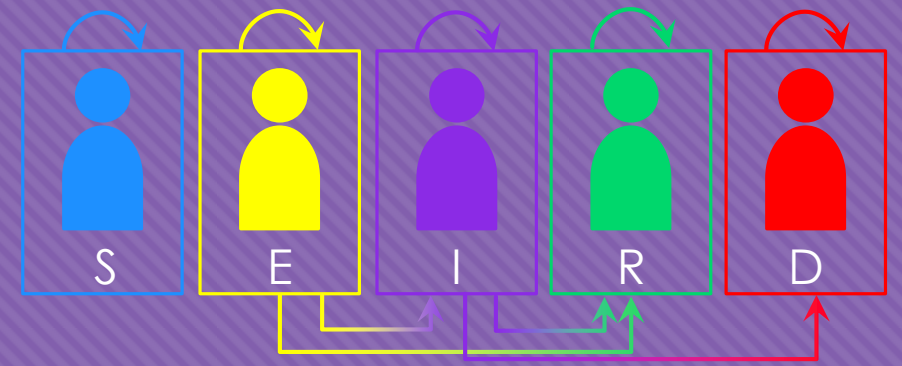


2.2 Contagion

$P \rightarrow \text{infect.h} = 0.1$; $P \rightarrow \text{infect.w} = 0.1$; $P \rightarrow \text{infect.hosp} = 0.1$



2.3 Patient's progress

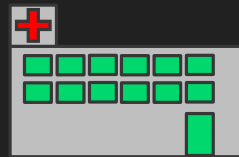


$P \rightarrow \text{evolve}[\text{hospital_info}][\text{initial_state}][\text{final_state}]$

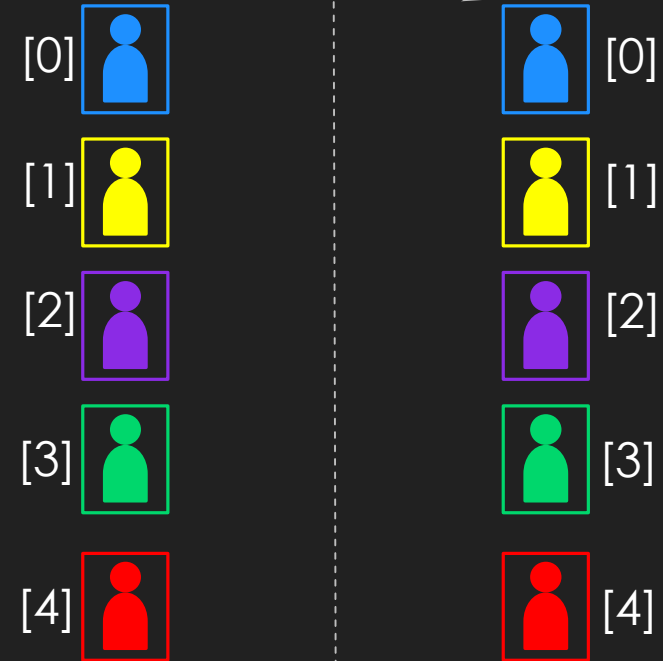
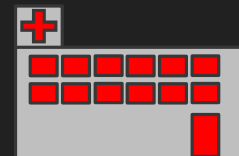
[0] → hasn't gone to the hospital.



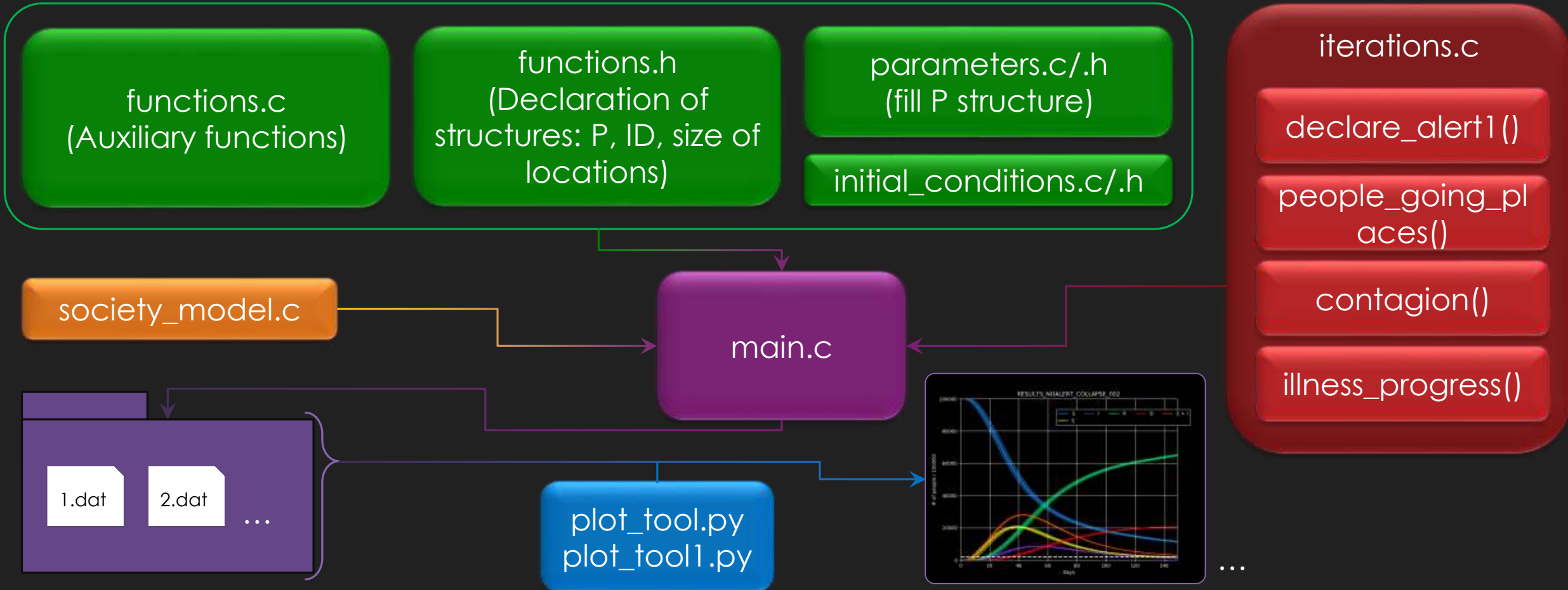
[1] → has gone to the hospital and it was not collapsed.



[2] → has gone to the hospital and it was collapsed.



2.4 Flow-chart



3. Results

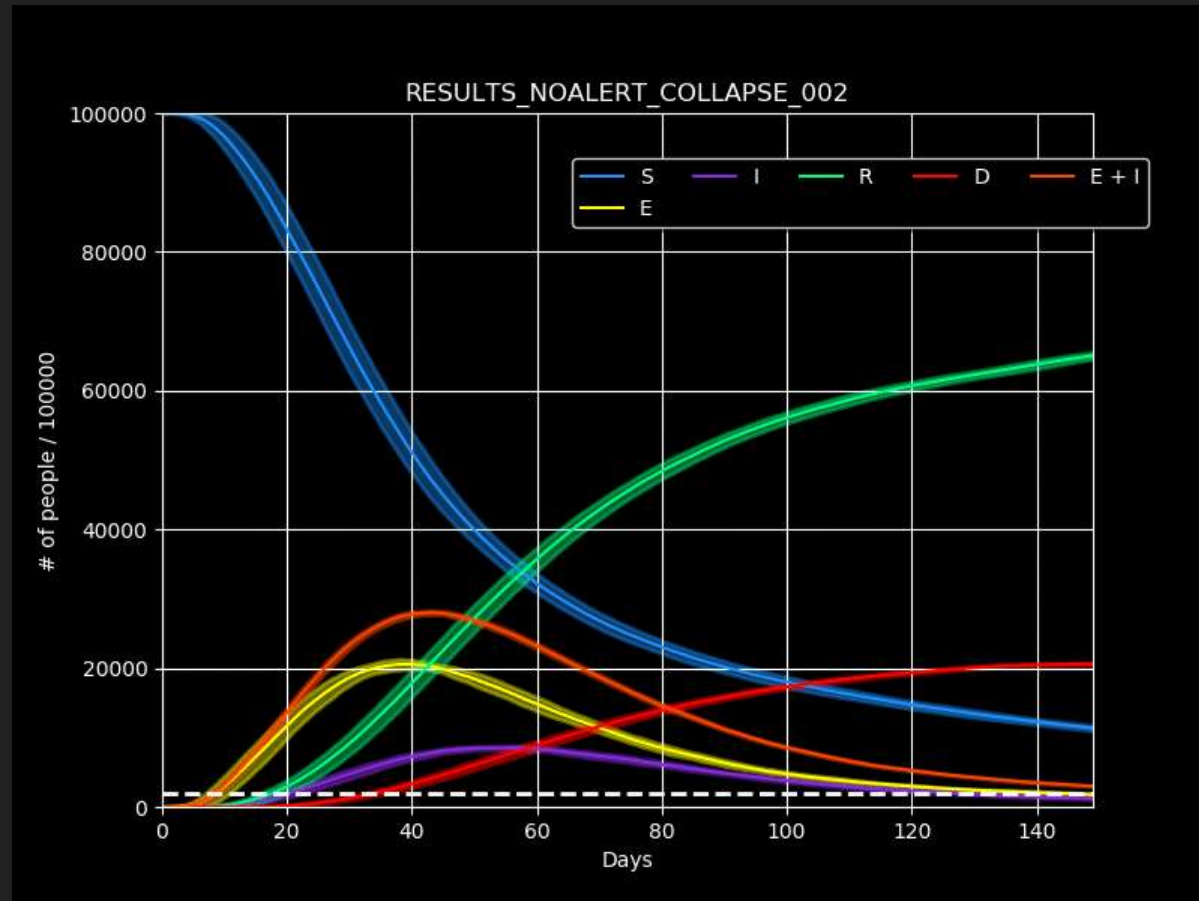
$$N = 32500 \rightarrow 1hosp$$

$$t = 150 \text{ days}$$

NO ALERT

% hospital collapse =
[0.02, 0.03, 0.04, 0.05, 0.06, 0.08, 0.1]

$$N_{realizations} = 10$$



3. Results

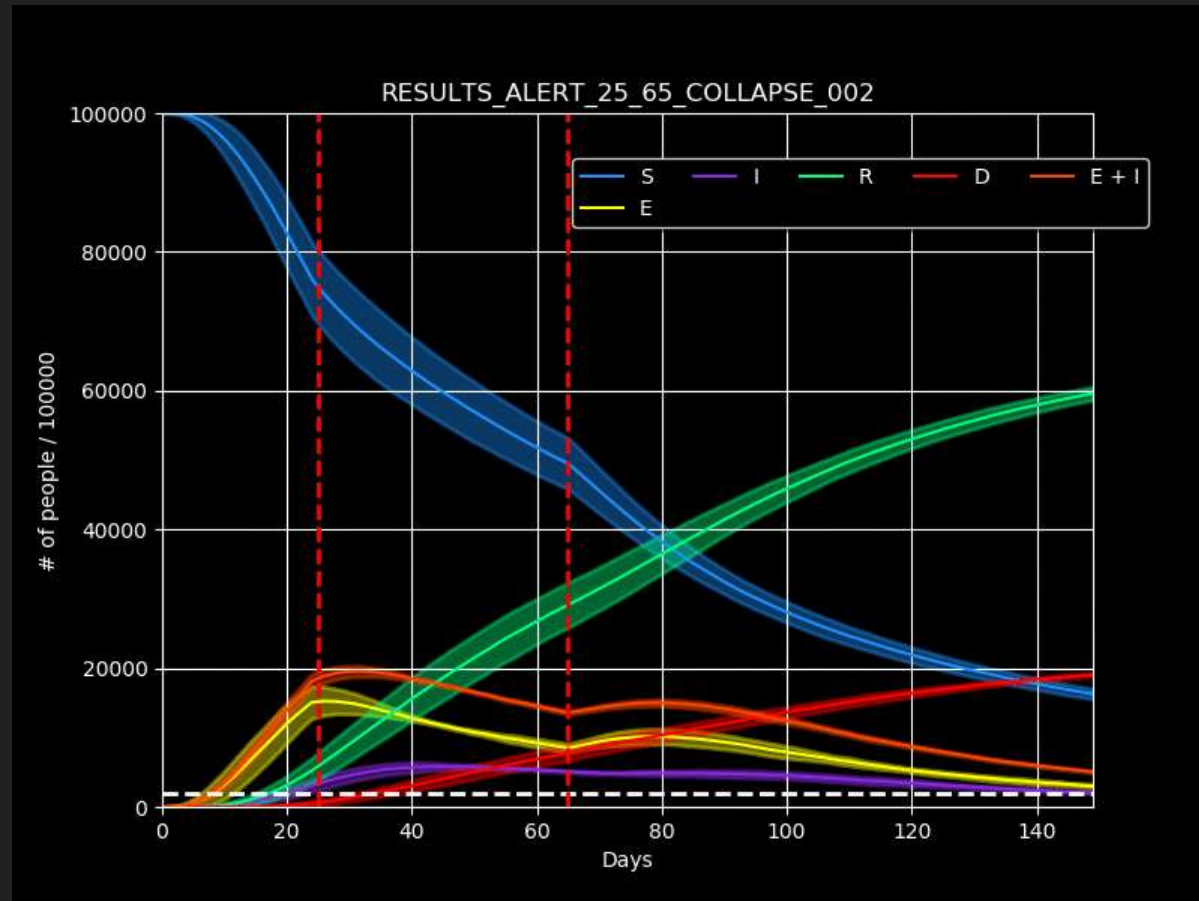
$$N = 32500 \rightarrow 1hosp$$

$$t = 150 \text{ days}$$

$$ALERT \in [25, 65]$$

$$\% \text{ hospital collapse} = [0.02, 0.03, 0.04, 0.05, 0.06, 0.08, 0.1]$$

$$N_{realizations} = 10$$



3. Results

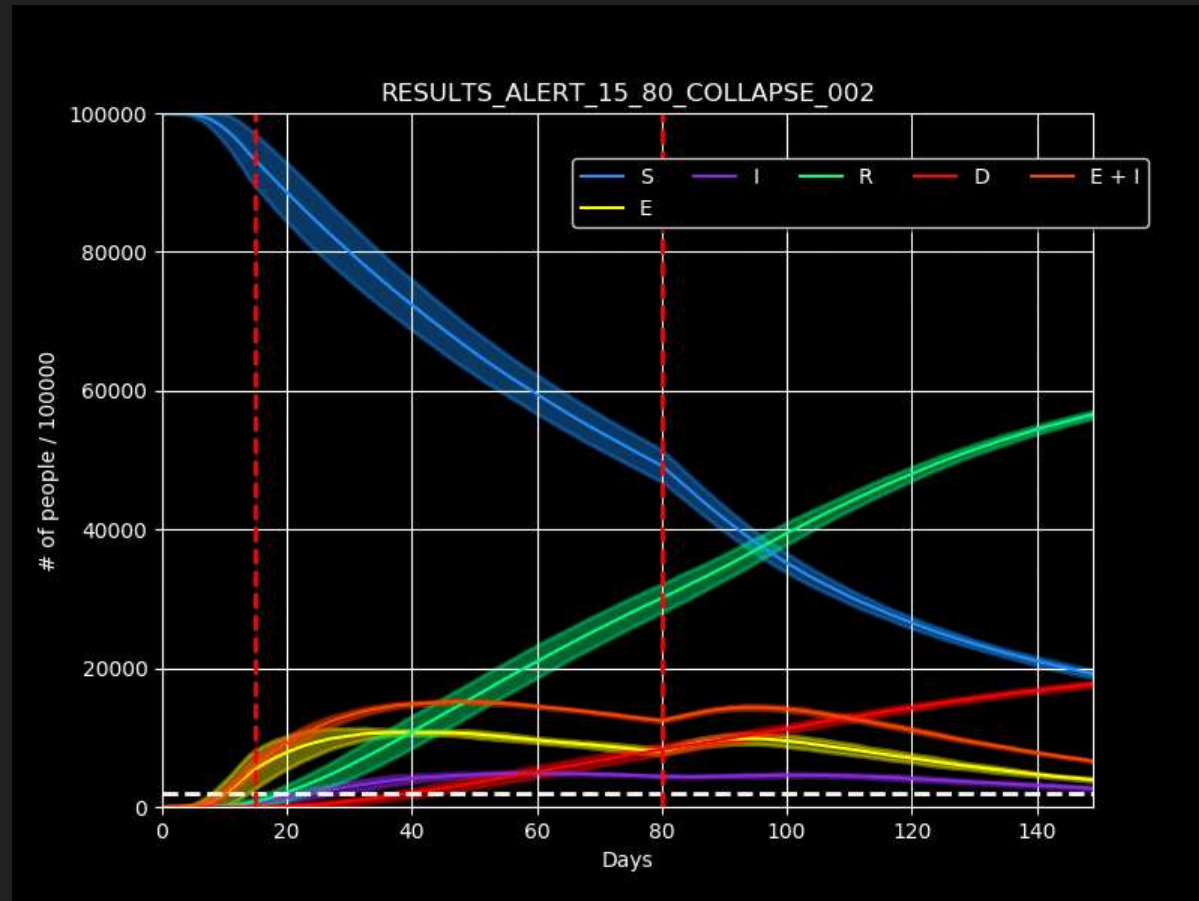
$$N = 32500 \rightarrow 1hosp$$

$$t = 150 \text{ days}$$

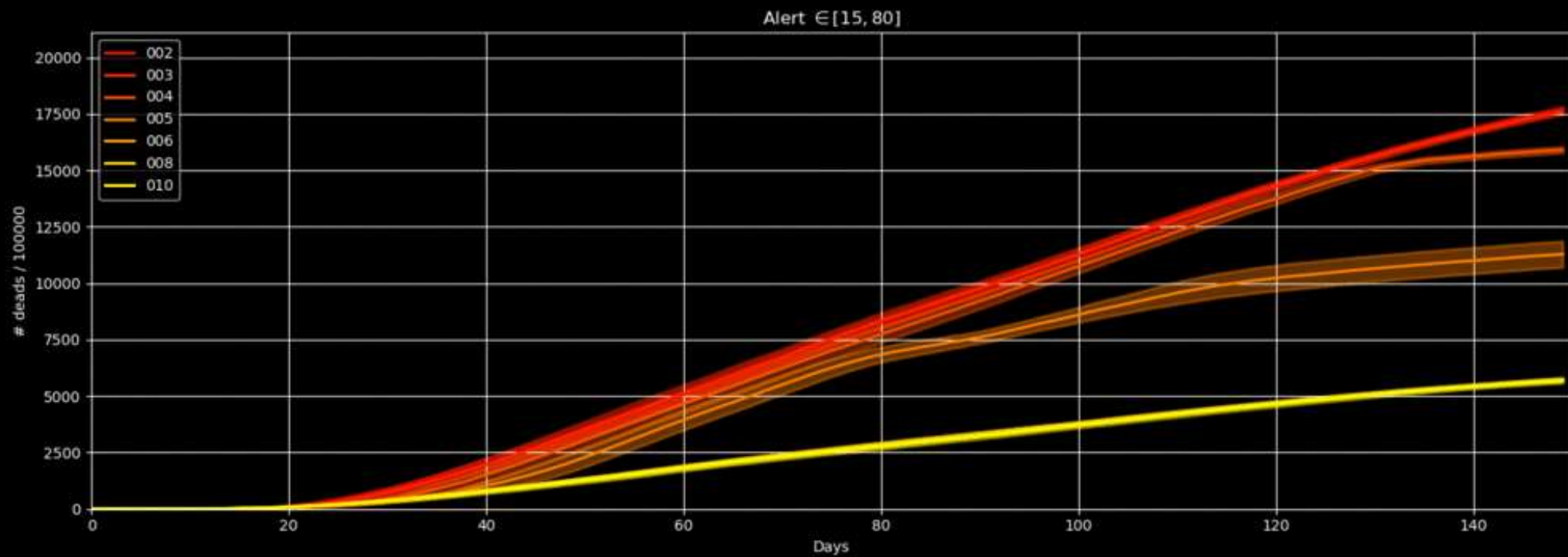
$$ALERT \in [15, 80]$$

$$\% \text{ hospital collapse} = [0.02, 0.03, 0.04, 0.05, 0.06, 0.08, 0.1]$$

$$N_{realizations} = 10$$



3. Results

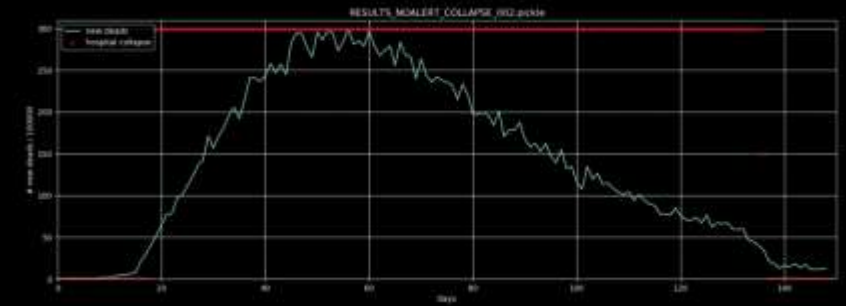
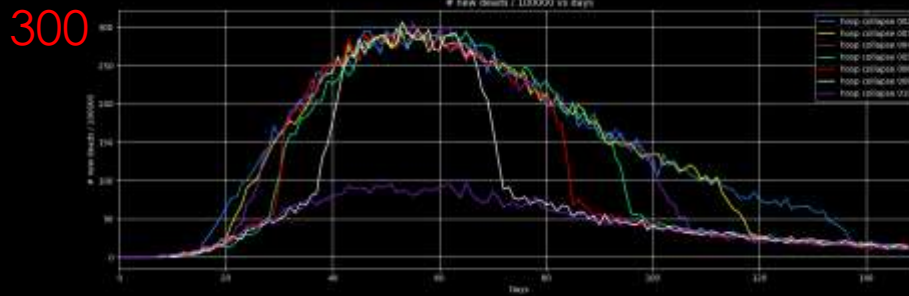


3. Results

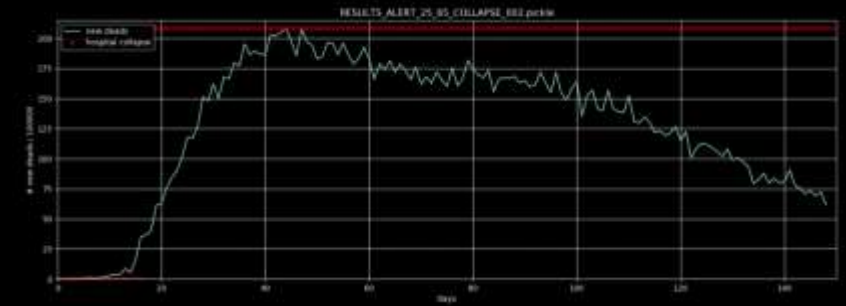
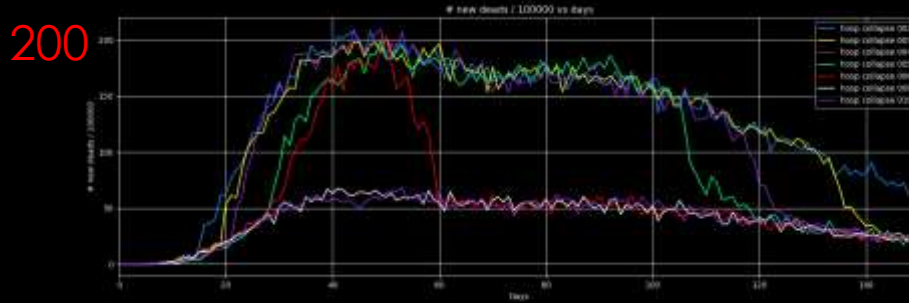


3. Results

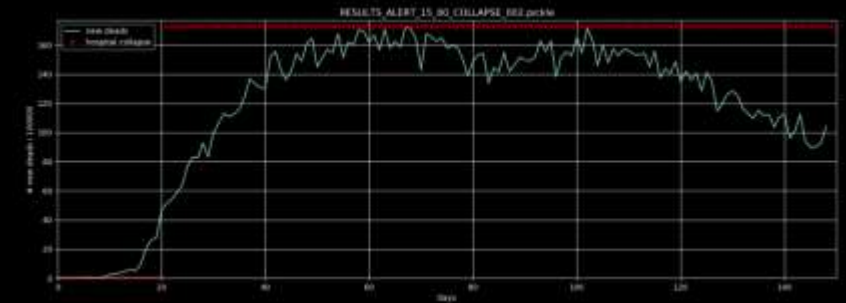
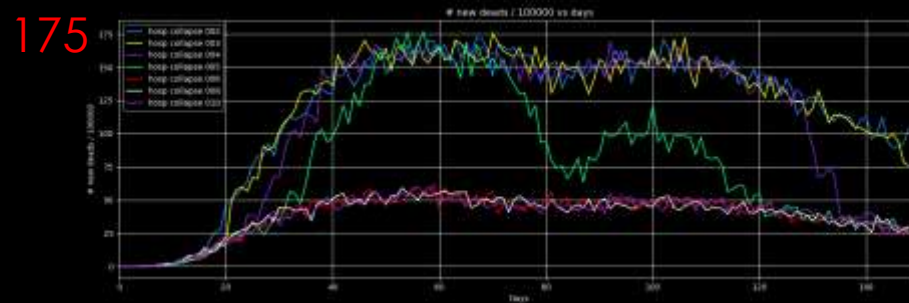
No alert:



Alert $\in [25, 65]$



Alert $\in [15, 80]$



4. Conclusions

- Two factors regulate the total number of deaths during a pandemic: the collapse threshold of hospitals and the social distancing measures regime.
- No matter how much the contagion curve flattens as long as hospitals are saturated. The total number of deaths may become similar in the end.
- Working with structures is indispensable when trying to model systems in which we deal with multiple elements to which we associate certain properties.